

Alejandro Sánchez Yalí

AI Full Stack Engineer



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SUMMARY

AI Full Stack Engineer with 8+ years designing and shipping web applications end to end and 3+ years building LLM-powered products. Architect and lead engineer of three deployed systems – an AI WhatsApp support platform, an AI-assisted recruitment platform and a surgical supply chain system – spanning Python/FastAPI backends, Next.js/TypeScript frontends, PostgreSQL and cloud deployment. M.Sc. in Mathematics and 11 years teaching mathematics and physics, which shows up in design docs and code review.

TECHNICAL SKILLS

- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL, Rust (personal projects)
- **Frontend:** React.js (7+ years), Next.js (6+ years), Tailwind CSS, Shadcn UI, Redux, Zustand, Zod, Effect, Three.js
- **Backend:** FastAPI, Django, Django REST Framework, Celery, Flask, NestJS, Node.js, Express, RESTful APIs, GraphQL, microservices
- **AI & LLMs:** LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, PyTorch, TensorFlow
- **Databases & Infra:** PostgreSQL, Redis, MongoDB, MySQL, Firebase, Docker, AWS, GCP, GitHub Actions, CI/CD
- **Engineering:** Clean architecture, DDD, SOLID, TDD, Vitest, Jest, React Testing Library, code review

EXPERIENCE

- | | |
|---|--|
| AI Specialist
<i>Lapzo</i> | Aug 2025 – Sep 2026
<i>Remote, Mexico</i> |
| <ul style="list-style-type: none">– Designed and built a Digital Professor that holds real-time conversations with learners and explains course content through interactive slides– Owned the architecture connecting LLM services, ElevenLabs voice synthesis and the real-time layer built with Next.js, NestJS and Firebase Realtime Database– Closed 57 tasks and 6 epics in the first nine months of 2026 with a single production bug, per the team's board– Built the AI content-generation pipelines (LangChain, LangGraph, n8n), with Celery workers running generation in the background | |
| Independent Software & AI Consultant
<i>Freelance</i> | Aug 2024 – Present
<i>Remote</i> |
| <ul style="list-style-type: none">– Delivered end-to-end web applications for business clients across both halves of the stack: React.js/Next.js/TypeScript on the front, Python/FastAPI/Django and Node.js/NestJS behind it– Lead engineer of Aluna (AI recruitment platform, see below) and builder of a multi-tenant AI customer service agent for CREARIA (Colombia) on WhatsApp Cloud API, LiteLLM, RAG and MCP with human escalation– Ran technical design sessions with clients and made the technology decisions for each project; code reviewer for a client's development team– Deployed on AWS (EC2, S3, Docker) with CI/CD pipelines and Stripe payment integrations | |
| Full Stack Engineer
<i>Monadical</i> | Apr 2021 – Aug 2024
<i>Remote, Canada/US</i> |
| <ul style="list-style-type: none">– Worked on client products such as ZoHuddle, a video conferencing app for scientific conferences (Neuromatch) that matches participants by mutual interests, built with Django, Django Channels (WebSockets), Twilio Video and React– Worked on OddSlingers, an in-browser poker platform for desktop and mobile built with Django and React on an event-stream architecture– Developed both frontend and backend, including reusable components and tests (Vitest, Jest, TDD)– Integrated AI/ML services and LLMs (OpenAI, LangChain, Whisper) into production web apps– Did code reviews on a distributed team across Canada, the US and Latin America | |
| Full Stack Engineer
<i>BCFort</i> | Aug 2018 – Oct 2020
<i>Medellín, Colombia</i> |

- Designed system architecture for blockchain platforms and built decentralised app frontends with React and Web3.js
- Developed analytics platforms with Python (Django) and Node.js, managing cloud infrastructure on AWS, GCP and IBM Cloud

Professor of Mathematics & Physics (part-time)
Universidad de Antioquia

Jan 2010 – Oct 2021
Medellín, Colombia

- Taught mathematics and physics; in the last year and a half also machine learning and data science with Python, NumPy, Pandas, TensorFlow and PyTorch
- Developed the AI/ML elective curriculum for the Mathematics Institute and mentored student research projects

FEATURED PROJECTS

- **Plixiq – AI Customer Support on WhatsApp**  – *Python, FastAPI, LiteLLM, PostgreSQL, Redis, Next.js, WebSocket*
 - Architect and lead engineer of a multi-tenant SaaS where AI agents answer routine customer inquiries, book appointments and escalate the rest to human operators with full conversation context
 - Modular monolith (14 components plus a shared kernel) with boundaries enforced in CI; LiteLLM layer with a primary and fallback provider per agent, so an outage becomes a fallback
- **VitaStock – Surgical Supply Chain for Clinics**  – *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway*
 - Architect and lead engineer of a platform commissioned by a private surgical clinic in Colombia – 12 modules and 5 user roles designed to replace spreadsheet-based inventory and procurement; delivered, with rollout at the clinic pending
 - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex
- **Aluna – AI Recruitment Platform for Staffing Agencies**  – *Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
 - Lead engineer of an AI recruitment platform for staffing agencies, built in 3 months – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
 - Durable Inngest analysis pipeline (9 checkpointed steps, 3 retries) where deduplication runs before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call; one screening engine serves WhatsApp, web chat and email

OPEN SOURCE

- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- **kিপux**  – Collaborative-finance app in Rust (Axum, sqlx, PostgreSQL, HTMX, SSE, PWA) with Claude-vision receipt parsing. Open source (MIT), work in progress
- 493 GitHub stars across own open-source projects – top 4.8% of ranked GitHub profiles (GitRanks)

TALKS & WRITING

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024
- **Building Plixiq: a Multi-Tenant WhatsApp AI Support Platform** – architecture case study (2026)

EDUCATION

M.Sc. in Mathematics
Universidad de Antioquia

Aug 2010 - Sep 2013
Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics
Universidad de Antioquia

Jan 2004 - Oct 2009
Medellín, Colombia